AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS IX EXAMINATION

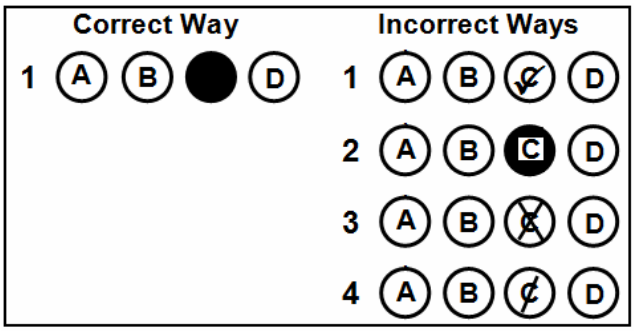
MAY 2014

Biology Paper I

Time: 35 minutes Marks: 25

INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 25 only.
4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.

1. A group of researchers worked on the availability of a less expensive fuel for cars. They came up with the discovery of a gas produced by micro-organisms. To which of the following branches of biology is their research related?

   A. Pharmacology
   B. Biotechnology
   C. Entomology
   D. Taxonomy

2. A student working on World Heart Day found out the average age at which most people are diagnosed with heart disease.

   The study shows the link of biology with

   A. mathematics.
   B. physics.
   C. economics.
   D. geography.

3. The diversity of living organisms present on the earth benefits us in all of the following ways EXCEPT

   A. maintaining ecosystems.
   B. recycling nutrients.
   C. reducing drug variety.
   D. producing more food.

4. In which of the following ways do fungi resemble plants?

   A. They are heterotrophs.
   B. They possess cell wall.
   C. They are prokaryotes.
   D. They release digestive enzymes.

5. According to Five Kingdom Classification System bacteria are not placed in Kingdom Plantae because

   A. their cell wall contains chitin.
   B. they lack a nuclear envelope.
   C. they are multicellular.
   D. they prepare their own food.

6. Which of the following CORRECTLY identifies the organelle shown in the diagram?

   A. Mitochondria
   B. Endoplasmic reticulum
   C. Ribosome
   D. Chloroplast



7. Lysosomes are most abundant in those cells which perform phagocytic activity. Which of the following cells has more lysosomes?

A. Nerve cell
B. Epithelial cell
C. Islets of Langerhans
D. White blood cell

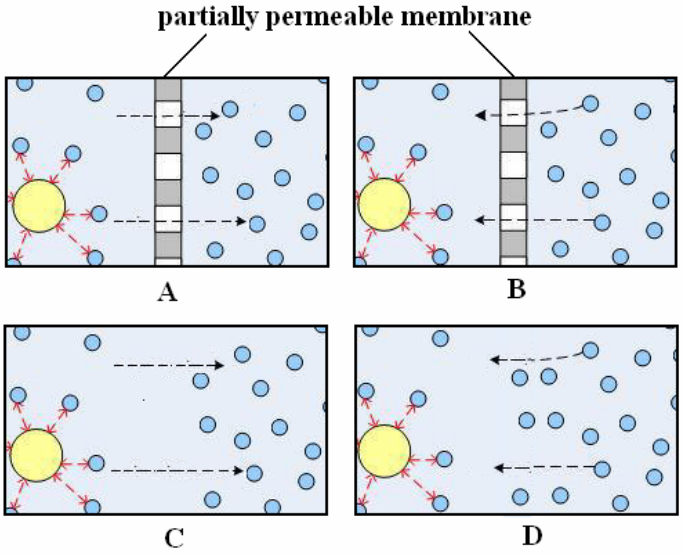
8. A type of animal tissue has the following characteristics.

i. It is found around kidneys.
ii. It contains fat droplets.
iii. It prevents heat loss.

The given description identifies

A. cartilage.
B. blood.
C. adipose tissue.
D. fibrous connective tissue.

9. Which of the following diagrams best represents the movement of molecules in osmosis?



10. Quantitative observations are considered more reliable than qualitative observations as quantitative can be

A. recorded.
B. measured.
C. tested.
D. questioned.

11. ‘When heating a test tube, always point the mouth of the test tube away from you.’

The above statement represents

A. an observation.
B. a hypothesis.
C. a safety measure.
D. a conclusion.

12. Two chromosomes together make up a homologous pair where both chromosomes have

A. different genes at different positions.
B. different genes at the same position.
C. same genes at different positions.
D. same genes at the same position.

13. Which of the following processes involves meiosis?

A. Increase in height
B. Repair of a wound
C. Growth of hair
D. Formation of sperm cells

14. Which phases of the cell cycle are collectively called interphase?

A. G_1, G_2 and M
B. G_1, G_2 and S
C. S, G_2 and M
D. G_1, M and S

15. During the development of brain, our body creates millions of cells, more than required. Those cells which do not form synaptic connections undergo death so that the remaining cells can function well. This is an example of

A. necrosis.
B. plasmolysis.
C. synapsis.
D. apoptosis.

16. Which of the following is an example of anabolism?

A. Conversion of proteins into amino acids
B. Increase in mass of the muscles
C. Burning down of fat deposits
D. Oxidation of organic substances

17. ‘Amylase breaks down starch and lipase breaks down lipids.’

The above statement about enzymes points out that enzymes are

A. very specific in nature.
B. required in small amounts.
C. sensitive to high temperature.
D. dependent on co-factor.

18. Which of the following features is NOT an adaptation of leaves to perform photosynthesis?

A. Larger number of stomata
B. Waxy cuticle on the leaf
C. Large leaf surface area
D. High concentration of chlorophyll

19. During strenuous exercises muscle cells carry out anaerobic respiration to meet energy demand. Afterwards, during rest, the lactic acid produced is

A. transported to the liver.
B. given off through lungs.
C. removed from the body in urine.
D. stored in the muscles permanently.

20. Look at the given food items carefully.



The food items shown above are main sources of

A. proteins.
B. fibre.
C. fats.
D. carbohydrates.

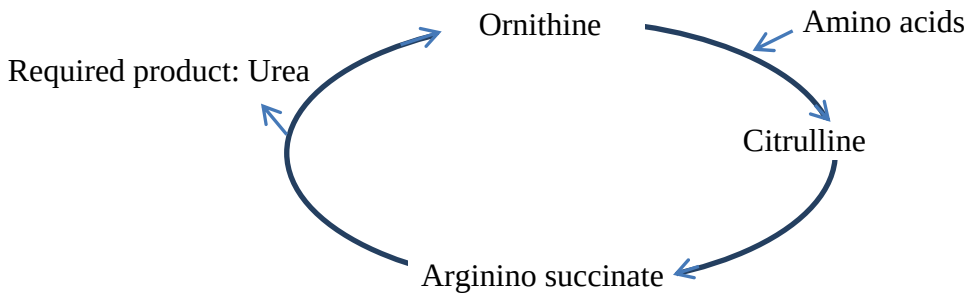
21. The table below shows estimated energy requirements (in kilocalories) according to age, gender and activity.

Gender	Age (years)	Activity Level		
		Sedentary	Moderately Active	Active
Male	4-8 (young)	1400	1400 – 1600	1600 – 2000
	31-50 (adult)	2200	2400 – 2600	2800 – 3000

The information in the given table implies that

A. adult males need to do more work than younger males.
B. adult males should take more carbohydrates than younger males.
C. the energy requirement of all males is the same.
D. males need to burn more calories than females.

22. The diagram shows a biochemical cycle.



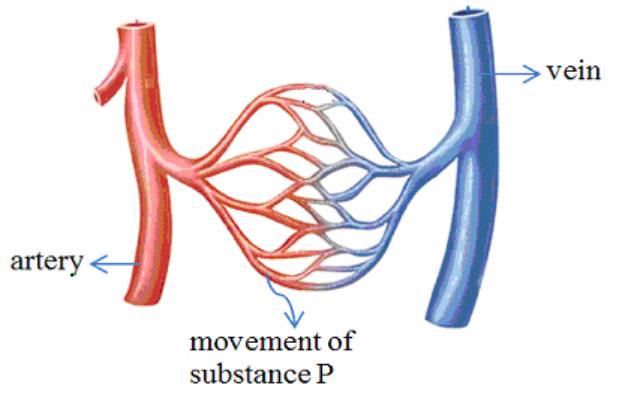
Looking at the required product of the cycle identify where does the given cycle take place in the human body?

A. Stomach
B. Villi
C. Liver
D. Pancreas

23. Which of the following statements is TRUE about guard cells? They

A. are rod-shaped cells.
B. do not have chloroplast.
C. are part of the epidermal layer.
D. become flaccid when they get water.

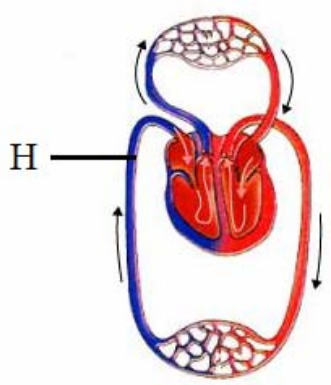
24. The diagram shows transfer of material through blood capillaries.



Which of the following CORRECTLY identifies substance P?

A. Glucose
B. Carbon dioxide
C. Red blood cells
D. Urea

25. The diagram shows double circulation in a mammal.



Which of the following CORRECTLY identifies blood vessel H?

A. Dorsal aorta
B. Inferior vena cava
C. Femoral artery
D. Pulmonary vein

END OF PAPER

Please use this page for rough work